DO NOW — RECALL

LMC instructions and data

Name: _____ Class: _____ Date: _____

1. What does direct addressing [30] retrieve?

2. What is the accumulator?

3. What is a mnemonic?

4. Search for 19 in sorted [1,4,7,10,13,16,19,22,25]. Show midpoint indexes checked.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

The data at memory address 30.

2. Model answer

A CPU register holding an arithmetic/intermediate value.

3. Model answer

A readable abbreviation of an operation code.

4. Model answer

low0 high8 mid4→13, low5 high8 mid6→19; found index 6.

Vocabulary — know these terms

LMC — Little Man Computer teaching instruction set.

ACC — Accumulator holding current working value.

mailbox — Numbered LMC memory location.

STA — Store accumulator to memory.

LDA — Load accumulator from memory.

DAT — Declare/initialise a data location.

NEW LEARNING

Read, annotate and explain

Know LMC's parts

LMC uses numbered mailboxes to hold instructions and data, and an accumulator for working values.

Trace instructions in order

First INP enters 4 into ACC, then STA FIRST copies 4 to the data mailbox before another input overwrites ACC.

Produce the result

Second INP makes ACC=7; ADD FIRST adds stored 4 to make 11, and OUT prints 11. HLT ends the program.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Write and trace an LMC program adding inputs 3 and 5.

Predict

Store the first input 3 into NUM so the second input will not overwrite it.

Explain

Read the second input 5, add NUM to get 8 and output the result.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Explain what INP, OUT, LDA and STA do in LMC.

Question 2

Trace INP; STA X; INP; ADD X; OUT; HLT; X DAT 0 using inputs 6 and 8. State each change and output.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Write an OCR LMC program that takes two inputs from 0 to 999, with the first greater than or equal to the second. Subtract the second from the first and output the non-negative result. Trace inputs 9 and 2.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

In LMC, STA X replaces ACC with the stored value of X.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.